

Correlation Polytope

$$\mathcal{P}_{\text{corr}}^{(n)} = \text{conv} \{ x x^T : x \in \{-1, 1\}^n \} \subseteq \mathbb{R}^{n^2}$$

$$y \in \mathcal{P}_{\text{corr}}^{(n)} \Leftrightarrow y = \sum \lambda_i x_i x_i^T, \quad x_i \in \{-1, 1\}^n.$$

$$y^0 \triangleq x^0 x^{0T} \leftarrow \text{extreme points of } \mathcal{P}_{\text{corr}}^{(n)}.$$

Thm: Ext complexity is at least 1.5^n .

A Special Face — 1/2

Vertices of P_{corr} : $y^B = \underline{x^B x^{B^T}}$

For a subset $A \subseteq [n]$: $\left(\sum_{i \in A} x_i - 1 \right)^2 \geq 0$

multiply out: $\left(\sum_{i \in A} x_i \right)^2 - 2 \sum_{i \in A} x_i + 1$

$$\boxed{\sum_{i,j \in A} x_i x_j - 2 \sum_{i \in A} x_i + 1}$$

$y^B \Rightarrow y_{ij} = x_i^B \cdot x_j^B$

For $y = x x^T$, $\forall x \in \{0,1\}^n$: $\sum_{i,j \in A} y_{ij} - 2 \sum_{i \in A} y_{ii} + 1 \geq 0$ // This linear inequality is satisfied $\forall$ vertices y^B

$= \cancel{\sum_{i=j \in A} y_{ii}} + \sum_{i \neq j \in A} y_{ij} - 2 \sum_{i \in A} y_{ii} + 1 \geq 0$

Thus: $\sum_{i \neq j \in A} y_{ij} - \sum_{i \in A} y_{ii} + 1 \geq 0$ valid $\forall y \in P_{\text{corr}}$.

A Special Face — 2/2

For $y = x^A$

$$\sum_{i \neq j \in A} y_{ij} - \sum_{i \in A} y_{ii} + 1 = \underbrace{\left(\sum_{i \in A} x_i - 1 \right)}_{\geq 0}$$

Which vertices $y^B = x^B x^B$ lie on this face?

$$A: \text{ all } y^B \text{ s.t. } \underline{|B \cap A| = 1.}$$

Call all these faces F_A , $A \subseteq [n]$.

Lemma: $\forall A \subseteq [n]$, $\exists$ linear constraint F_A that is valid for all of P_{arr} , and s.t. the ~~set~~ vertex $y^B \in F_A$ iff $|A \cap B| = 1$.

Road Map

we want to lower bound comm complexity
of FACE-VERTEX.

① We focus only on the subset of slack
matrix corresponding to faces $\underline{F_A}$ of
the form in prev. lemma.

② We make the problem even easier: We need Bob & Alice to
agree w/ help of prover when $FV(f, r) = 1 \rightarrow \boxed{FV(A, B)}$
Alice & Bob: $FV(A, B) = 1$?
 $\uparrow$
This happens when $|A \cap B| \neq 1$

$f = F_A$ $r = y^B$

Easier: declare B & A win if $|A \cap B| \geq 2$. In other words: Bob and Alice
only need to agree on 1's in the subset of $\text{supp}(S)$ that correspond
to $A \cap B = \emptyset$

Define:

$$\mathcal{D}(n) = \left\{ (A, B) \in 2^{[n]} \times 2^{[n]} : A \cap B = \emptyset \right\}$$

↑

These are the "1's" in the subset of
the slack matrix, that Bob & Alice must agree
on with help of the "primes".

②

We show that an Ext form Q for P_{cor} , that has
 r neg. implies, that there is a covering of $\mathcal{D}(n)$
of same size.